Edexcel A Level Physics

Your notes

Transverse & Longitudinal Waves

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- Properties of Waves
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- Core Practical 6: Investigating the Speed of Sound

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Properties of Waves

- Waves are generated by **oscillating sources**
  - These oscillations travel **away** from the source
- Oscillations can propagate through a **medium** (e.g. air, water) or in a **vacuum** (i.e. no particles), depending on the wave type

Wave Features

- In order to describe the properties of travelling waves, the following keywords need to be defined:
  - **Wavelength** λ (m) is the distance between a point on a wave and the same point on the next cycle of the wave, e.g. two crests, or two troughs
  - **Amplitude** A (m) is the magnitude of the maximum displacement reached by an oscillation in the wave
  - **Period** T (s) is the time taken for one complete oscillation at one point on the wave
  - **Frequency** f (Hz) is the number of complete wave cycles per second
  - **Wave speed** c (m s^{-1}) is the rate of movement of the wave

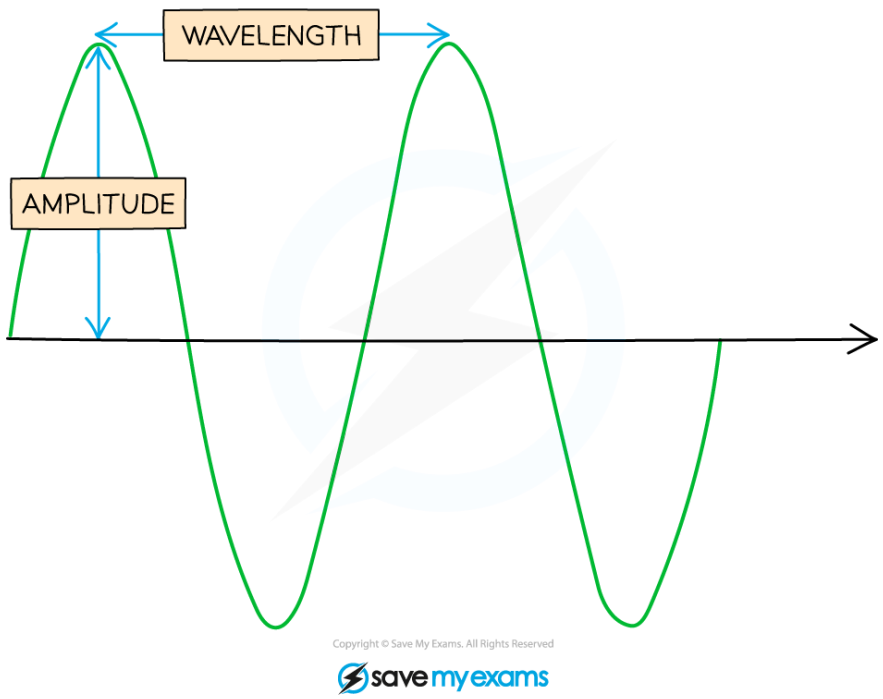
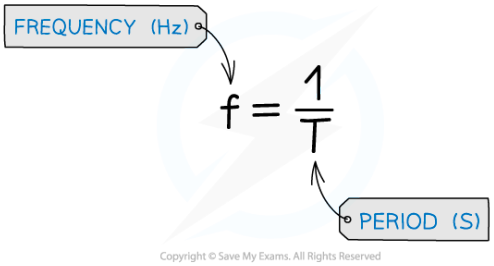


Diagram showing the amplitude and wavelength of a transverse wave

- The frequency *f* and the period *T* of a travelling wave are related to each other by the equation



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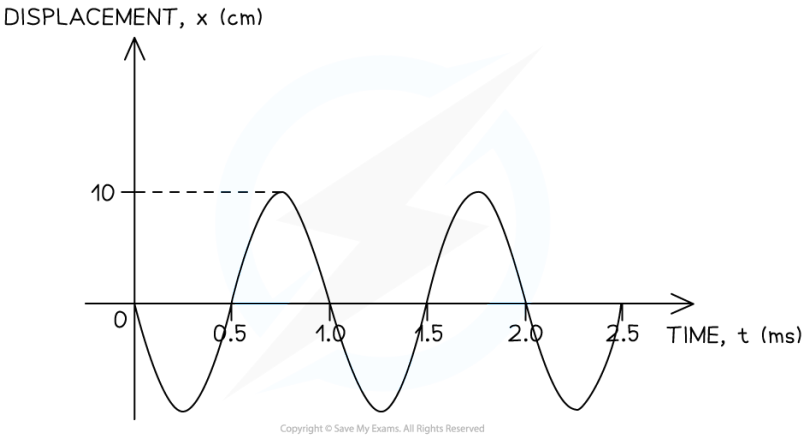
Worked Example

The graph below shows a travelling wave.



Determine:

(i) The amplitude *A* of the wave in metres (m)

(ii) The frequency *f* of the wave in hertz (Hz)

Answer:

(i) Identify the amplitude *A* of the wave on the graph

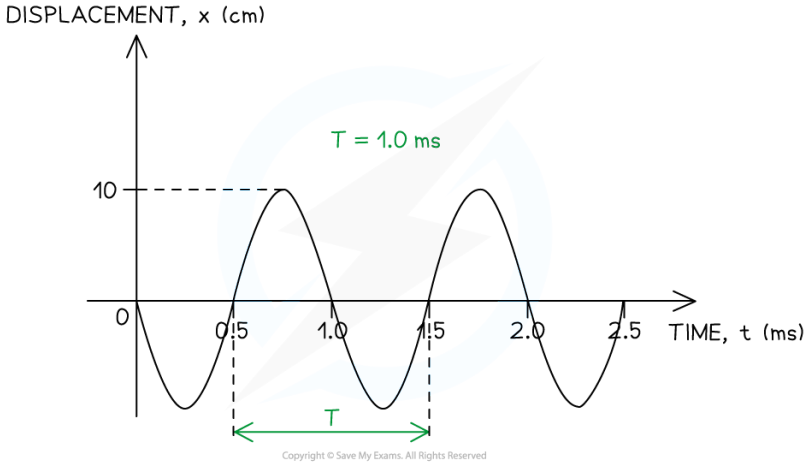
- The amplitude is defined as the maximum displacement from the equilibrium position ($x = 0$)
  The amplitude must be converted from centimetres (cm) into metres (m)

$$A = 0.1\ \text{m}$$

(ii) Calculate the frequency of the wave

Step 1: Identify the period *T* of the wave on the graph

- The period is defined as the time taken for one complete oscillation to occur



- The period must be converted from milliseconds (ms) into seconds (s)

$$T = 1 \times 10^{-3}\ \text{s}$$

Step 2: Write down the relationship between the frequency *f* and the period *T*

$$f = \frac{1}{T}$$

Step 3: Substitute the value of the period determined in Step 1

$$f = \frac{1}{1 \times 10^{-3}}$$

$$f = 1000\ \text{Hz}$$

Examiner Tips and Tricks

Every question about waves uses the vocabulary on this page. Many questions start by asking you to define one of them.

So your job is to learn the definitions to the point where you have them memorised.

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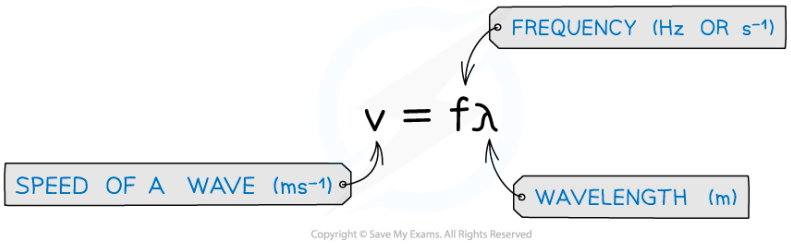
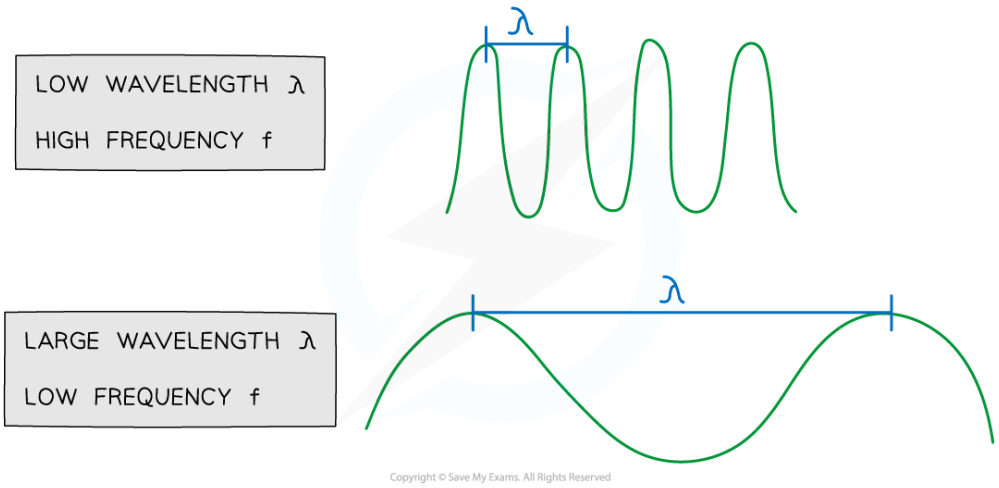
The Wave Equation

The Wave Equation

- This equation links wave speed, frequency and wavelength



- Where:
  - v = velocity of the wave (m s^{-1})
  - f = frequency of the wave (Hz)
  - λ = wavelength (m)
- The wave equation tells us that for a wave of constant speed:
  - As the wavelength **increases**, the frequency **decreases**
  - As the wavelength **decreases**, the frequency **increases**



The relationship between frequency and wavelength of a wave

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Worked Example

A travelling wave has a period of 1.0 μs and travels at a velocity of 100 cm s^{-1}. Calculate the wavelength of the wave. Give your answer in metres (m).

Answer:

Step 1: Write down the known quantities

- Period, $T = 1.0\ \mu s = 1.0 \times 10^{-6}$ s
- Velocity, $c = 100$ cm $s^{-1} = 1.0$ m s^{-1}

Note the conversions:

- The period must be converted from microseconds (μs) into seconds (s)
- The velocity must be converted from cm s^{-1} into m s^{-1}

Step 2: Write down the relationship between the frequency *f* and the period *T*

$$f = \frac{1}{T}$$

Step 3: Substitute the value of the period into the above equation to calculate the frequency

$$f = \frac{1}{1 \times 10^{-6}}$$

$$f = 1.0 \times 10^{6}\ \text{Hz}$$

Step 4: Write down the wave equation

$$c = f\lambda$$

Step 5: Rearrange the wave equation to calculate the wavelength λ

$$\lambda = \frac{c}{f}$$

Step 6: Substitute the numbers into the above equation

$$\lambda = \frac{1.0}{1 \times 10^{6}}$$

$$\lambda = 1 \times 10^{-6}\ \text{m}$$

Examiner Tips and Tricks

This is an important equation that comes up in many other topics. Get really familiar with using it and rearranging it.

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Be ready to use **prefixes** with values, for example, nanometres (nm = m × 10^{-9}) or MHz (MHz = Hz × 10^{6}).

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Longitudinal Waves

Longitudinal Waves

- A longitudinal wave is one where the particles oscillate **parallel** to the;
  - Propagation of the wave
  - Direction of energy transfer
- Longitudinal waves show areas of
  - **High pressure,** called **compressions**
  - **Low pressure**, called **rarefactions**

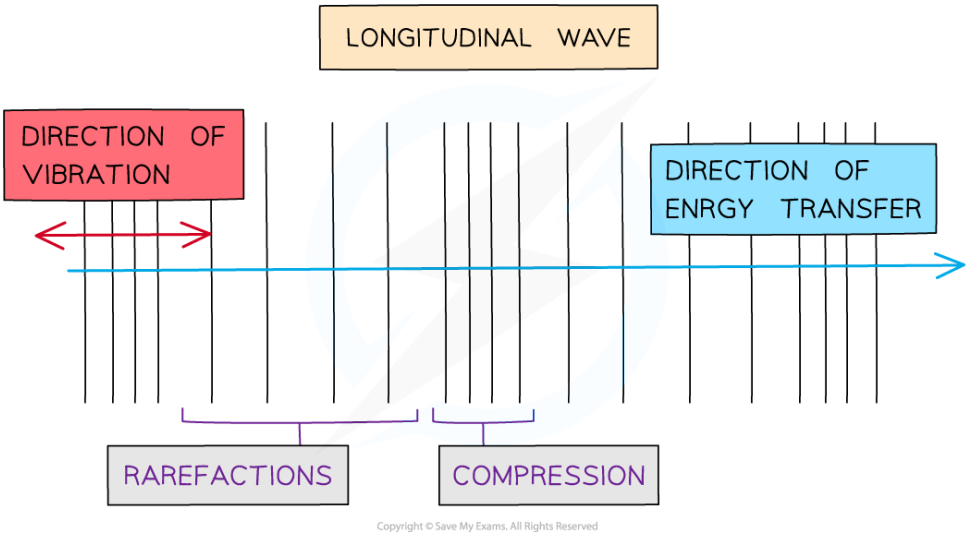


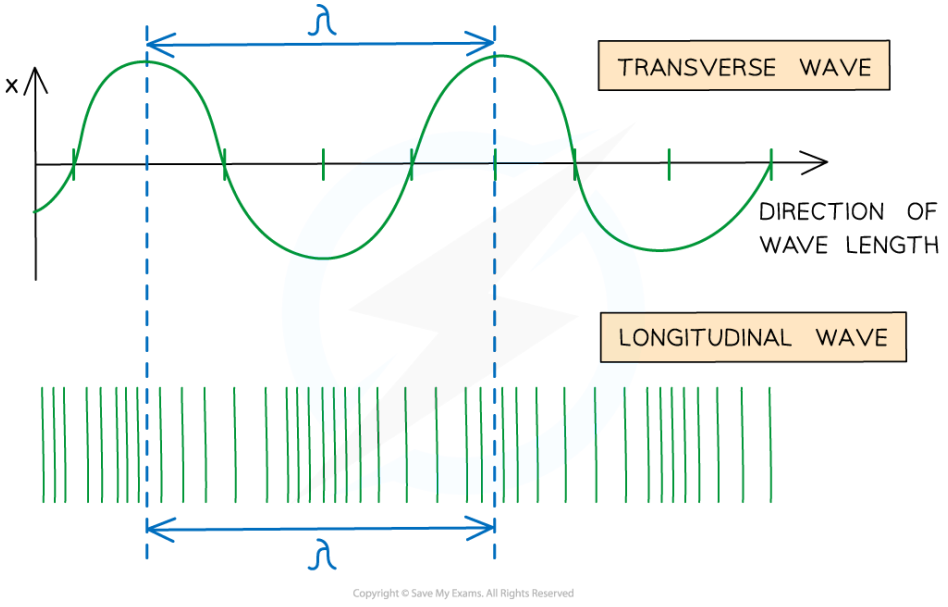
Diagram of a longitudinal wave

- Examples of longitudinal waves are:
  - Sound waves
  - Ultrasound waves
  - P-waves caused by earthquakes
- Longitudinal waves **cannot** be polarised

Labelling Longitudinal Waves

- You learned how to describe the properties of a wave, such as amplitude and wavelength at the start of this topic

- The diagram shows a wavelength on a longitudinal wave



Wavelength is shown on a longitudinal wave

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Examiner Tips and Tricks

Questions about longitudinal waves typically start by asking for a definition, so be ready with a statement about areas of high and low pressure and the keywords compression and rarefaction.

Be careful with graphs of waves and don't assume a sinusoidal-shaped graph represents a transverse wave. Longitudinal waves can also look sinusoidal when plotted on a graph - make sure you read the question and look for whether the wave travels **parallel** (longitudinal) or **perpendicular** (transverse) to the direction of travel to confirm which type of wave it is.

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Transverse Waves

Transverse Waves

- A transverse wave is one where the particles oscillate **perpendicular** to the direction of the
  - Propagation of the wave
  - Direction of energy transfer
- Transverse waves show areas of **crests** (peaks) and **troughs**

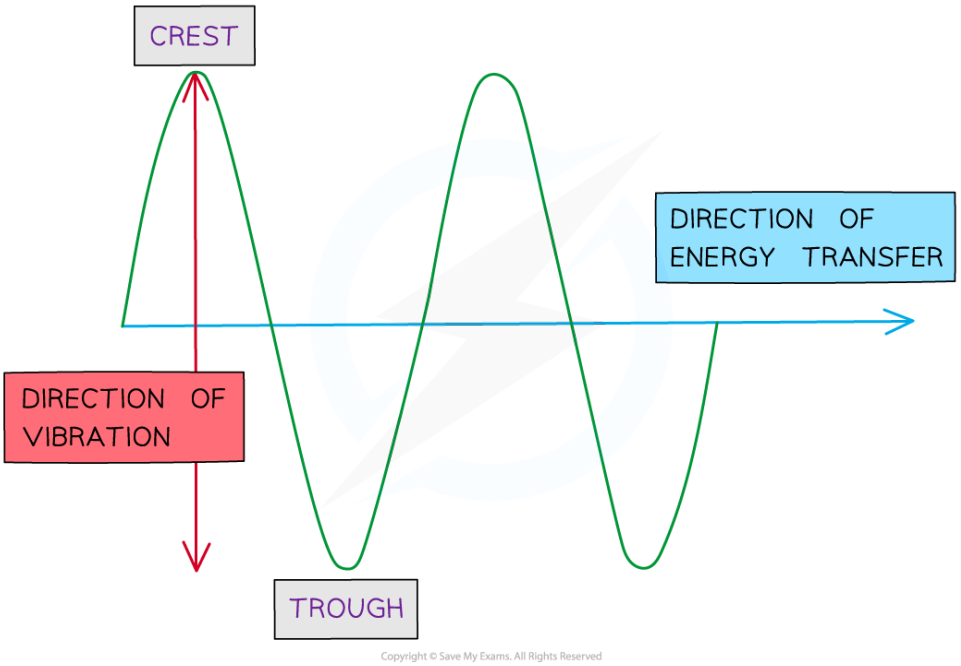

Diagram of a transverse wave

- Examples of transverse waves are:
  - Electromagnetic waves e.g. radio, visible light, UV
  - Vibrations on a guitar string
  - Waves on a rope or slinky
- Transverse waves can be polarised

Examiner Tips and Tricks

Questions about transverse waves typically start by asking for a definition, so be ready with a statement about **vibrations** or **oscillations** being perpendicular to the travel of the wave.

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Graphs of Transverse & Longitudinal Waves

Graphs of Transverse Waves

- There are two common graphs transverse waves;
  - Displacement against distance
  - Displacement against time
- These are:
  - **Similar** because they produce a **sinusoidal** shaped curve
  - **Different** because displacement against distance is showing displacement of a point on the wave, but displacement against time is showing the wave itself moving along a line
- On the **displacement-distance graph**:
  - Movement upwards from the centre line is given a positive sign and movement downwards a negative
  - The amplitude and wavelength can be found as shown below

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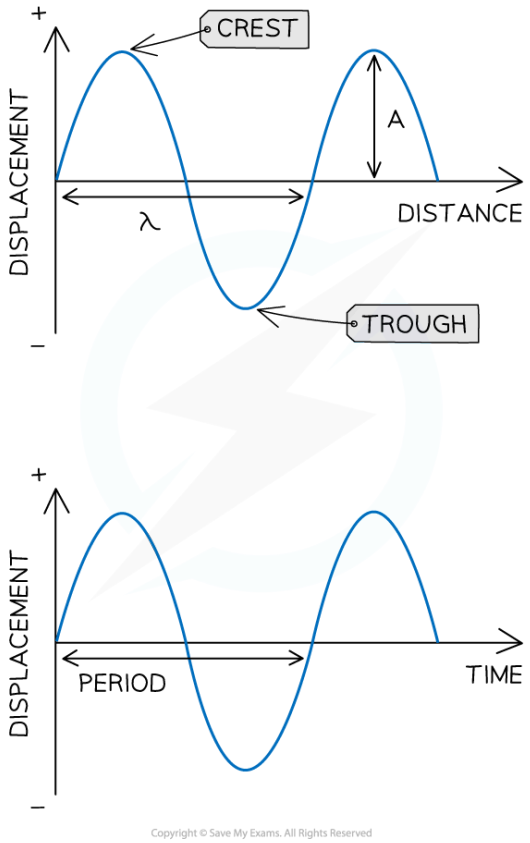
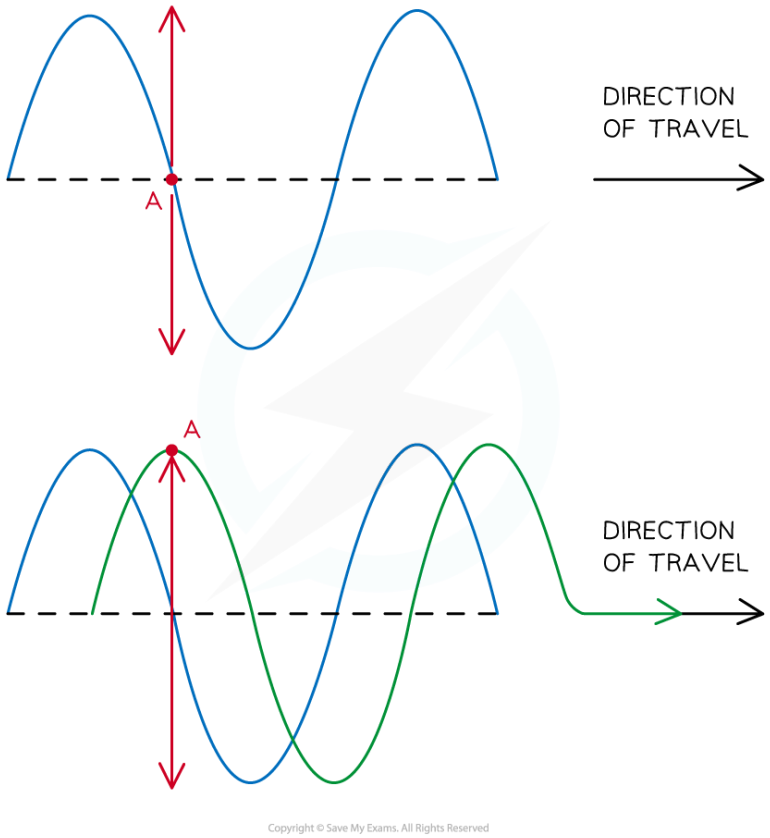
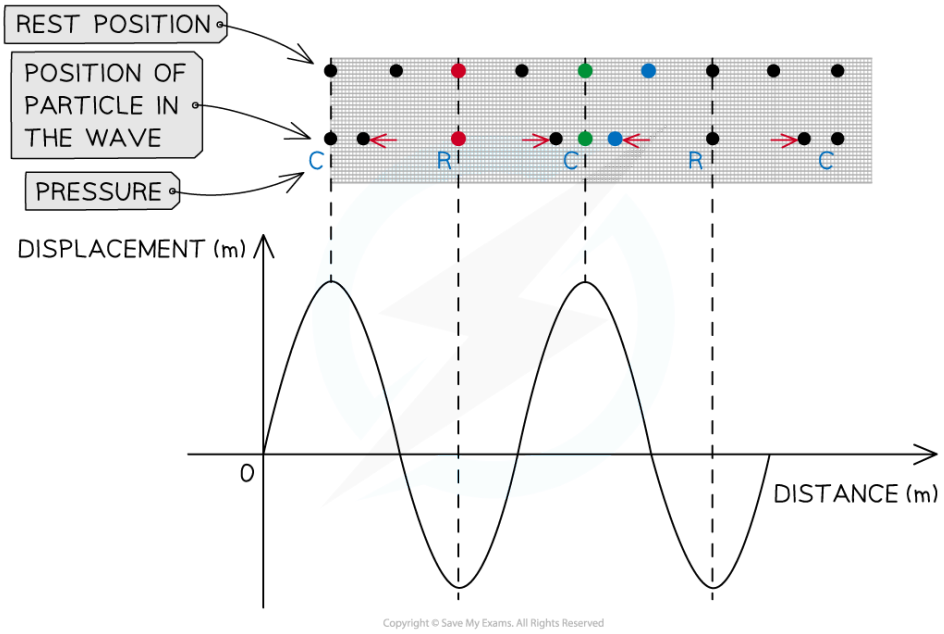


- On the **displacement-time graph:**
  - The time period can be taken directly as shown
  - This means that frequency can be found indirectly as f = 1/T
- To determine the next position of a point on the wave
  - Sketch the full wave after time has passed by looking at the direction of travel
  - Each point oscillates perpendicular to the wave, so remains on the normal line wherever the wave intersects, this is shown in red below



Graphs of Longitudinal Waves

- Plotting displacement against distance also produces a sinusoidal shaped graph
  - This can be used to show where the compressions and rarefactions will be found



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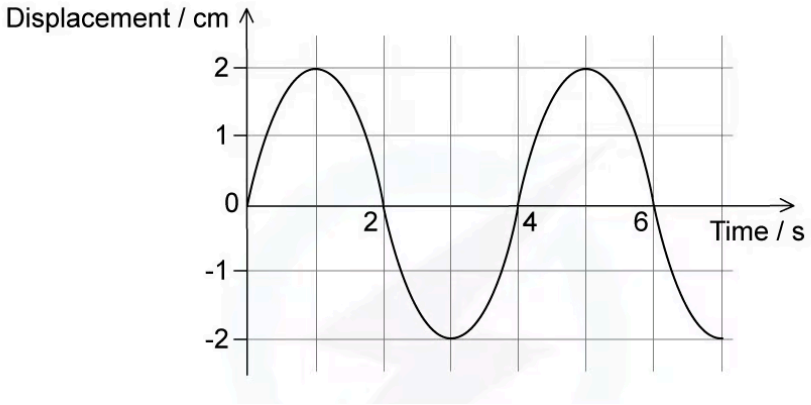
Worked Example

The graph shows how the displacement of a particle in a wave varies with time.



Which statement is correct?

A. The wave has an amplitude of 2 cm and could be either transverse or longitudinal.

B. The wave has an amplitude of 2 cm and has a time period of 6 s.

C. The wave has an amplitude of 4 cm and has a time period of 4 s.

D. The wave has an amplitude of 4 cm and must be transverse.

Answer: A

THE WAVES AMPLITUDE IS THE DISPLACEMENT FROM THE EQUILIBRIUM POSITION

FROM THE GRAPH, THIS IS 2 cm

THE GRAPH IS DISPLACEMENT AGAINST TIME, NOT DISPLACEMENT AGAINST DIRECTION OF WAVE TRAVEL

THEREFORE, THE WAVE COULD BE EITHER TRANSVERSE OR LONGITUDINAL

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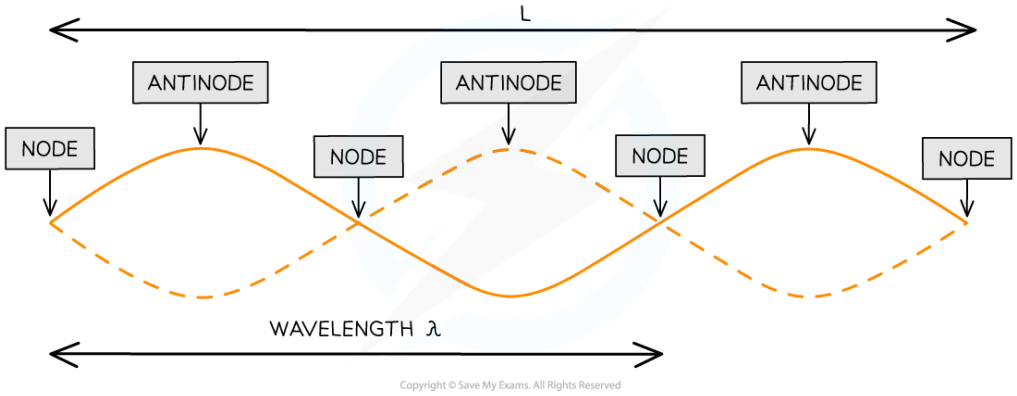
Graphs of Stationary Waves

- Stationary waves occur when a wave is reflected with a 180° phase difference, creating a wave with a series of nodes and antinodes
  - Stationary waves can be transverse or longitudinal

- They are represented graphically in the same way as progressive (travelling) waves
- Graphs of standing waves can also be used to determine the position of nodes and antinodes



- *L* is the length of the string
- 1 wavelength λ is only a portion of the length of the string

Examiner Tips and Tricks

Both transverse and longitudinal waves can look like transverse waves when plotted on a graph - make sure you read the question and look for whether the wave travels **parallel** (longitudinal) or **perpendicular** (transverse) to the direction of travel to confirm which type of wave it is.

Your notes

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Core Practical 6: Investigating the Speed of Sound

Aim of the Experiment

- To measure the speed of sound in air using an oscilloscope and a signal generator

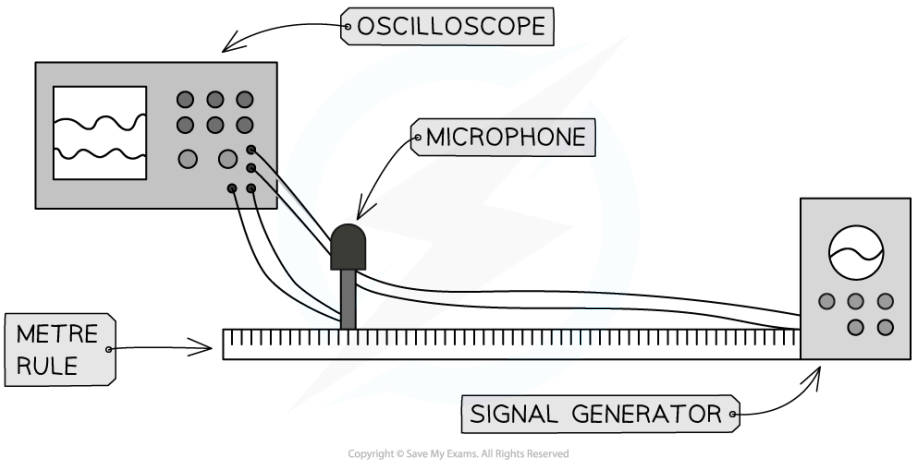
Variables

- Independent variable = Distance
- Dependent variable = Phase of received signals
- Control variables:
  - Same location to carry out the experiment
  - For **each set of readings**, the same frequency of sound

Equipment

- Signal generator with loudspeaker
- Oscilloscope with 2-beam facility
- Microphone
- 2 metre rulers or 1 measuring tape of at least 2 m length
- Connecting leads

Method



1. Connect the microphone and signal generator to an oscilloscope, and set up the signal generator about 50 cm from the microphone

2. Set the signal to about 4 kHz
3. The oscilloscope should trigger when the microphone detects a sound, adjust the time base so that the signal from the generator and the microphone can be on the screen with about three cycles visible
4. Adjust the separation so a trough on the upper trace coincides with a peak on the lower trace (this makes judging the point where the waves coincide easier)
5. Record the distance between the microphone and signal generator (call this distance 1, d_1)
6. Move the microphone further away, watch the traces on the screen
7. When the next trough and peak coincide record the new distance (call this distance 2, d_2)
8. Repeat steps 6 and 7 as many times as possible in the available space (numbering the distances as required)
9. Calculate the mean wavelength of the sound
10. Using the oscilloscope trace find the frequency of the sound
11. Reduce the frequency to around 2 kHz (or half of the original value) and repeat steps 4–10.

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Table of Results:

TIME PERIOD /μ s	FREQUENCY /kHz	d_1/m	d_2/m	d_3/m

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Analysis of Results

- The speed of sound can be calculated using the equation:

$$v = f\lambda$$

- Frequency is found from the time base of the oscilloscope by using

$$f = \frac{1}{T}$$

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Evaluating the Experiment

Systematic Errors:

- Ensure the scale of the time base is accounted for correctly
  - The scale is likely to be small (e.g. milliseconds) so ensure this is taken into account when calculating frequency
- Use the oscilloscope signal trace to find frequency to avoid relying on the dial of the signal generator

Random errors:

- Random errors in taking measurements can be reduced by doing repeat readings and taking an average
- The time interval is small so make the distance between the microphone and signal generator as large as is practical

Safety Considerations

- The voltage and current are low, so normal care with electrical equipment is sufficient (including checking the leads for any signs of damage)
- Keep sound at a normal listening volume to avoid damage to hearing

Examiner Tips and Tricks

When you are answering questions about methods to measure waves, the question could ask you to comment on the accuracy of the measurements

When measuring the speed of sound, this experiment is very accurate because the timing is done automatically so reaction time is not a factor

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